# From Equity Pairs to Options Relative Value

Does learning improve capital allocation?

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**Abstract**

We extend equity-pairs trading to relative implied volatility in approximately 30-day, 25-delta puts on related equity underlyings. A screened opportunity rule generates trades; learning then determines their entry size. In a simulated $10 million account over January 2022–August 2025, constant full sizing achieves annualized excess Sharpe 1.055, compared with 1.263 for decision-tree states and cross-entropy sizing. The observed gain of 0.208 has a conditional 95% interval of $[-0.103, 0.424]$. The learner admits 110 trades compared with 75 under constant full sizing. Performance varies across years, and previously inspected history, different training and evaluation conventions, and simplified execution limit the conclusions. The comparison supports positive historical profit above cash and a higher point estimate for learned sizing, without establishing a repeatable advantage from learning.

## 1 Introduction

Suppose AMD and NVDA face similar news, but AMD options become unusually expensive relative to NVDA options. Selling the richer options and buying the cheaper ones expresses a view about the relative price of volatility. Daily stock hedges reduce immediate directional exposure, and initial vega matching reduces sensitivity to a common small change in implied volatility. The trade still carries risk: related stocks can experience different events, and option premiums may correctly reflect those differences.

Our starting point is Ning and Lee's equity-pairs framework [1], which separates constructing a spread from learning how to trade it. We carry that distinction into options. First, we ask whether a screened relative-volatility rule produces useful historical trades under explicit contract and funding assumptions. We then ask the central question: *does learning improve how capital is allocated to those opportunities?* A profitable signal and a useful allocation policy are separate findings.

After explaining the motivating paper and our adaptation, we define the trade, monthly training procedure and tree/CEM sizing method. We then compare constant and learned sizing in the same funded account and examine the learner's annual results. The comparison uses the retained configuration and saved historical backtests.

## 2 Motivating paper and adaptation

### What the paper does

Ning and Lee [1] study how to construct and trade mean-reverting equity spreads. At time $t$, let $S_{1,t}$ and $S_{2,t}$ denote the two stock prices. They form the spread $X_t = S_{1,t} - BS_{2,t}$, where $B$ is the hedge coefficient, and choose $B$ by grid search to minimize empirical mean reversion time: the average duration from a significant local extreme to the next mean crossing. Formation does not fit an Ornstein–Uhlenbeck (OU) mean-reversion model.

For trading, they use tabular Q-learning. The state encodes recent spread moves by direction and magnitude; available actions buy, hold or sell a unit, subject to the current position. The policy enters a long spread and later closes it, and does not initiate a short spread. Its reward favors buying below and selling above the equilibrium mean, with a transaction-cost term $c|A|$, where $c$ is the cost coefficient and $|A|$ is the magnitude of the trading action. Q-values combine immediate reward with the estimated value of later decisions.

Because the reward requires a true equilibrium mean, the agent trains on 10,000 one-year simulated OU paths before application to historical stocks. The market study forms ten US pairs in 2022 and trades them in 2023. Published daily Sharpe ratios exceed the Distance Method and OU benchmarks for all ten pairs. Those daily figures are not directly comparable with our annualized options Sharpe ratios.

### Criticisms that motivate our design

Three issues limit interpretation. **Short history:** one trading year provides limited evidence across regimes. **Costs and scale:** the reward defines a cost term, but its numerical implementation and reconciliation to net performance are not clearly documented. This is not evidence that costs were omitted. Higher turnover raises total costs; larger or faster orders can also increase per-unit slippage and market impact. A constant coefficient does not establish scalability. **Simulation transfer:** formation is empirical, but policy training still uses specified mean-reverting dynamics and a known equilibrium. Real relationships can shift, so extensive synthetic experience does not establish market transfer.

### What we borrow and what we change

We borrow the separation of opportunity construction from trading decisions, and the question of whether learning improves on a simple rule. We change the instrument from equity-price spreads to option-implied-volatility spreads, using actual contracts, initial vega matching and daily stock delta hedges. Completed historical outcomes replace a known synthetic equilibrium for training. We make simulated costs explicit and compare multiple trading years.

The next section defines relative-IV opportunities, option contracts and hedging through expiry. Our regression tree groups completed historical opportunities; cross-entropy search chooses an entry size for each group. This is supervised partitioning followed by sizing search, rather than sequential Q-learning. Comparing with constant sizing measures the additional historical value of learned allocation.

# 3 Our strategy and evaluation

## 3.1 From a volatility signal to a hedged trade

Implied volatility (IV) reproduces an option quote under a pricing model; realized volatility (RV) measures stock-price variation. RV inputs use trailing data available at the signal date. We trade puts near 25 delta and 30 days to expiry (P25), retained after exploratory research and fixed across sizing policies. We do not claim that this segment is optimal.

For underlyings $A$ and $B$, let $IV_{A,t}$ and $IV_{B,t}$ be selected-option IVs on signal date $t$. Rank their difference $D_t = IV_{A,t} - IV_{B,t}$ within the 252-session history ending on that date, with at least 63 observations. Above the 80th percentile, sell A's relatively rich put and buy B's; at or below the 20th percentile, reverse the direction. The percentile compares the pair with its own history, allowing for persistent IV differences.

At monthly formation, estimate beta of A's returns on B's and correlation using the preceding 126 sessions. Require beta between 0.85 and 1.15 and correlation of at least 0.5. Also require positive average net profit per initial vega for the proposed direction in full-size standalone simulations completed within the preceding 91 and 365 calendar days, with at least 21 and 63 outcomes, respectively. These simulations use training costs, can overlap and need not have been executed by the funded account. An unfinished trade's eventual profit cannot inform a current decision.

Quantities approximately match initial vega: option-value sensitivity to a one-percentage-point IV increase. Table 1 illustrates the entry positions at delta $-0.25$ per share and 100 shares per contract.

| Position | Contracts | Vega per IV point | Initial stock hedge |
|---|---|---|---|
| Buy A puts | +100 | +\$450 | Buy 2,500 shares |
| Sell B puts | −150 | −\$450 | Short 3,750 shares |

Table 1: Illustrative vega matching: A contracts have \$4.50 vega each; B contracts have \$3.00.

Hold each option to expiry and adjust stock delta hedges daily. Profit combines premiums, settlement payoffs and realized hedge cash flows, after costs and financing. The IV-rank signal is a hypothesis about relative option richness; initial vega matching and temporary IV convergence do not guarantee profitable expiry outcomes.

### 3.2 Simulating trades in a funded portfolio

Each policy uses one account initially funded with $10 million. Historical quotes, prices, rates and dividends come through WRDS. We inherit an earlier 345-pair roster and formation calendar; 125 pairs with 161 equity and ETF underlyings supply prepared candidates. Of these, 46 pairs generate 608 screened proposals over 2018–2025, including 312 in the later period. The universe is conditioned on prior research and available data, rather than independently selected for this comparison.

Contracts are selected on the signal date and traded, if eligible, on the next session. Whole-contract quantities must leave at most 10% initial vega mismatch. Accepted positions share cash and capital; stock hedges in the same name are netted. Existing trades can therefore prevent new entries. All policies receive identical proposals, quote checks and admission rules; Appendix A defines the contract and risk checks.

Option trades fill at midpoint with an execution charge equivalent to 0.10 IV percentage point of traded vega. For trading day $t$, let $\mathcal{F}_t$ be paid option fills, $\Delta n_{j,t}$ contracts traded in option $j$, and $V_{j,t}$ one contract's dollar vega per IV point. For stock $s$, let $P_{s,t}$ be price and $\Delta h_{s,t}$ the change in net hedge shares. Dollar execution cost is

$$C_t^{\text{exec}} = \sum_{j \in \mathcal{F}_t} |\Delta n_{j,t}|(0.10 V_{j,t} + 0.03) + 0.0001 \sum_s P_{s,t} |\Delta h_{s,t}|.$$

The constants are $0.03 per option contract and one basis point of stock turnover. Expiry settlement carries no option execution charge.

Let $B_{t-1}$ be prior cash, $H_{t-1}^-$ short-stock notional and $\Delta\tau_t$ elapsed calendar days divided by 365. Dollar funding charges are

$$C_t^{\text{fund}} = [0.003 H_{t-1}^- + 0.01 \max(-B_{t-1}, 0)]\Delta\tau_t.$$

These charge 0.3% annual short-stock borrow and a 1% surcharge on negative cash. Cash interest is $I_t = r_{t-1} B_{t-1} \Delta\tau_t$, where $r_{t-1}$ is the prior annual benchmark rate as a decimal. Negative cash pays both that rate and the surcharge. Corporate-event exits, missing-quote handling and exercise approximations are summarized in Appendix A.

Because unused cash earns interest, we evaluate trading value above an all-cash benchmark, including premiums, settlement, hedges, dividends, costs and financing. Total account growth alone does not measure trading performance.

### 3.3 Training, selection and testing

Each month, we train a new tree and size map using historical examples from the preceding 18 months, then apply that model to the next month's entry decisions. Figure 1 separates this rolling process from the choice of model specification. Training and application use successive slices of the same market history.

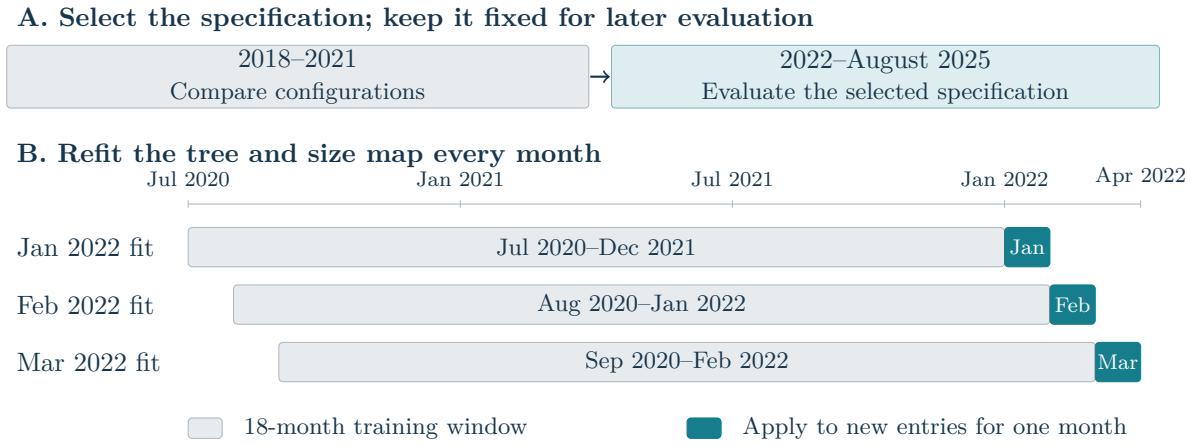



Figure 1: Two decisions, one historical dataset. The upper panel selects the 27-input, 18-month, 50/75/100% specification using earlier performance. The lower panel illustrates monthly refitting during evaluation. A training example is included only if its signal date falls within the window and its trade outcome is completed before that month's fit.

We retain the configuration with the highest annualized daily excess Sharpe on 2018–2021 among 33 existing configurations spanning training-history, feature and sizing-action choices. The selected configuration uses 27 inputs, an 18-month window and 50/75/100% actions. Candidate identifiers break score ties; its selection score is 0.575 in a $345 million research account. This configuration is then replayed at $10 million; the account assumptions are summarized in Appendix A.

Each monthly fit uses candidates that passed the entry screens on their signal dates, have valid simulated outcomes, and finished before fitting. These examples need not have been accepted by the funded account. Only signal dates within the preceding 18 months enter the fit. Tree fitting and CEM use this same sample; the wider candidate history supplies the profitability screens.

The later comparison runs from 3 January 2022 to 29 August 2025. The model specification stays fixed while monthly refitting continues: a completed March trade can inform an April fit, but an unfinished outcome cannot. New signals stop on 30 June 2025. All accounts continue from 2018 with their own accumulated cash and holdings; the 2022 boundary is a reporting split, not an account reset.

Later history had already been examined during strategy development. Monthly fits respect time order, but this period is not independent of earlier choices of contracts, screens and universe. Training also uses $0.65 commission per option contract, compared with $0.03 for every policy in final evaluation. We therefore interpret the comparison as historical evidence.

We report annualized Sharpe of daily returns above the cash benchmark, profit above cash and cash-relative drawdown. Appendix B defines these metrics and the opening portfolios at the reporting boundary.

### 3.4 Learning an entry size: decision tree and CEM

The opportunity rule fixes the pair, direction and contracts. We learn only the entry size: 50%, 75% or 100% of a reference budget of \$450 vega per IV point on each leg, scaled by current account equity relative to its initial equity. Opportunities differ in their historical outcomes, so the learner groups them and assigns a common size to each group (Figure 2).

Each monthly regression tree receives 27 inputs: 15 numerical characteristics and 12 availability flags, covering relative IV, trailing realized variance, relatedness, maturity, the CBOE Volatility Index (VIX), costs and completed profitability. Appendix A.1 defines the inputs. It learns splits using historical full-size profit above cash as its target, splitting to reduce squared prediction error [3]. Each of its up to eight leaves is a state. The cross-entropy method (CEM) [2] then chooses one size for each state.

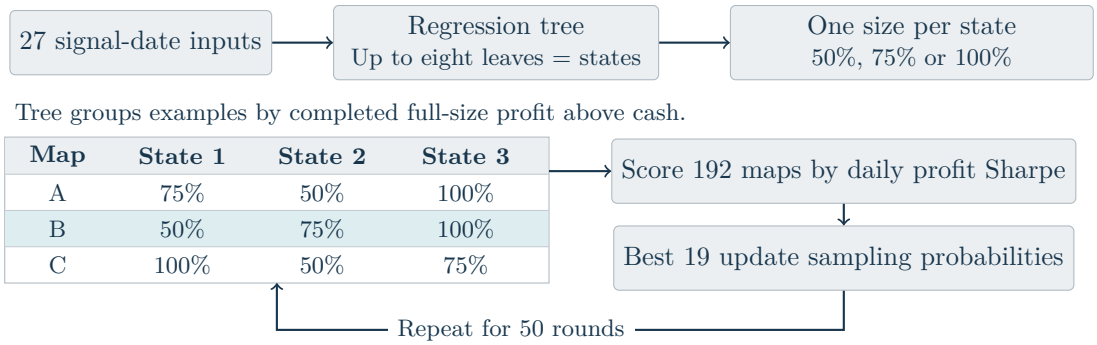



Figure 2: Tree partitioning followed by CEM sizing search. The three-state assignments illustrate candidate maps; actual trees have up to eight states. The retained map applies to the next month's new proposals.

#### How CEM scores a size map

A map $\boldsymbol{m}$ assigns size $a_\ell \in \{0.50, 0.75, 1.00\}$ to tree state $\ell$. For completed example $i$, $x_i$ is its input vector and $\ell(x_i)$ its state. Let $p_{i,d}(a)$ be profit above cash on trading day $d$ at size $a$. Each size uses its own rounded contracts and standalone cash and hedge flows, rather than scaling the full-size outcome. A common stored normalization cancels from the score. Sum paths and calculate

$$g_d(\boldsymbol{m}) = \sum_i p_{i,d}\Big(a_{\ell(x_i)}\Big), \qquad \widehat{S}_{\text{train}}(\boldsymbol{m}) = \sqrt{252}\,\frac{\overline{g(\boldsymbol{m})}}{s_{g(\boldsymbol{m})}}.$$

Here $g_d$ is combined daily profit; the numerator and denominator are its sample mean and standard deviation. The factor $\sqrt{252}$ annualizes using 252 trading days. The calendar retains overlapping trades and zero-profit sessions. This standalone-profit training score differs from funded-account Sharpe based on percentage returns and joint admission.

CEM updates size probabilities toward the best 19 of 192 maps, repeating for 50 rounds. Constant maps and the best map found so far are retained as anchors. A new proposal receives its tree state's requested size, subject to account admission. The search does not establish a global optimum.

# 4 Benchmark and learned-sizing results

## 4.1 Does learned entry sizing improve the benchmark?

We compare the tree/CEM policy with constant requests of 50%, 75% and 100% of the same reference budget. Each policy receives the same proposals and keeps the proposed direction and contracts.

Figure 3 shows when profit above cash accumulates. Most of the learner's final lead develops during 2022; both policies lose ground during 2024.

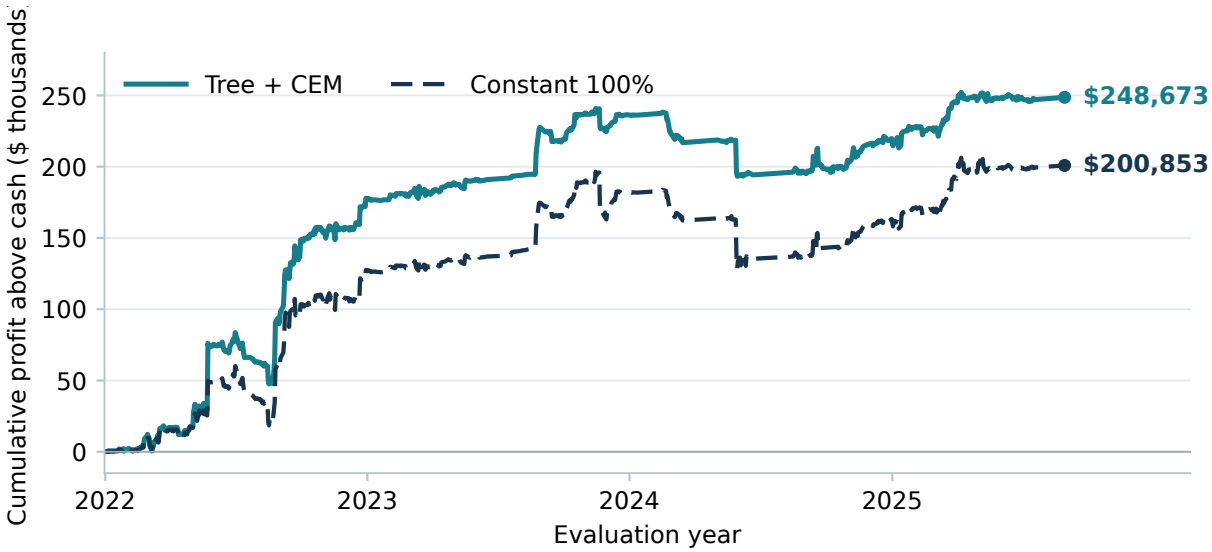



Figure 3: Cumulative profit above cash, 3 January 2022–29 August 2025. Curves are rebased at the 31 December 2021 close; the books retain carried positions. Endpoints match Table 2. The paired 95% interval for the Sharpe gain, $[-0.103, 0.424]$, includes zero.

Under the same proposals and funded-account rules, constant full sizing earns later excess Sharpe of 1.055 and profit above cash of $200,853, compared with 1.263 and $248,673 for the learner (Table 2). The observed gain is 0.208 Sharpe and approximately $47,800 additional profit. This portfolio-policy comparison includes changes in which proposals the account admits: 110 entries for the learner versus 75 for constant full sizing.

| Policy | Earlier Sharpe | Later Sharpe | Later entries | Profit above cash ($) | Relative drawdown (%) |
|---|---|---|---|---|---|
| Constant 50% | 0.437 | 1.007 | 151 | 204,913 | 0.735 |
| Constant 75% | 0.449 | 0.944 | 104 | 195,021 | 0.748 |
| Constant 100% | 0.478 | 1.055 | 75 | 200,853 | 0.646 |
| Tree + CEM | 0.573 | 1.263 | 110 | 248,673 | 0.495 |

Table 2: Earlier: 2018–2021 replay at $10 million; later: January 2022–August 2025. Entries, profit and maximum cash-relative drawdown refer to the later period. Earlier scores are $10 million replay scores, not the $345 million configuration-selection scores.

The paired circular-block bootstrap, using 126-session blocks and 1,000 replications, gives a 95% interval of $[-0.103, 0.424]$ for the Sharpe gain. It includes zero and omits search adjustment. The observed gain therefore does not establish a repeatable sizing improvement. It also does not isolate gains from sizing trades common to both policies.

The account earns annualized total return of 4.667%, while annualized growth relative to an all-cash account is 0.530%. Most total account growth therefore comes from cash interest. Trading is active—110 new entries—but its net contribution above cash is modest: $248,673 over the later period. Average reserve is 2.44% of equity and peak reserve 15.77%. Smaller requested sizes can pass stress and pair limits that reject full-size proposals, even when average account reserve is low. We use excess Sharpe and profit above cash to assess the strategy's added value, rather than attributing the full 4.667% account return to trading.

### 4.2 How does performance vary across years?

Table 3 breaks the learner's later performance into calendar periods. Its strongest annual cash-relative performance occurs in 2022, which also accounts for 66 of the 110 later entries. In 2024, the account earns a positive total return but loses relative to cash. The partial 2025 period is positive above cash again. Together with Figure 3, these results show that the aggregate profit reflects uneven performance across years.

| Year | Account return (%) | Above cash (pp) | Excess Sharpe | Entries |
|---|---|---|---|---|
| 2022 | 3.470 | 1.673 | 2.458 | 66 |
| 2023 | 5.288 | 0.432 | 1.492 | 19 |
| 2024 | 5.104 | −0.283 | −0.907 | 11 |
| 2025 (partial) | 3.207 | 0.169 | 1.092 | 14 |

Table 3: Tree/CEM annual funded results; 2025 ends on 29 August. Account and above-cash returns are unannualized for each displayed period, including partial 2025. Above-cash figures are return differences in percentage points; Sharpe is annualized.

## 5 Conclusion and limitations

We compare constant sizing with a tree/CEM policy that learns entry sizes from completed historical trades. Both policies use the same screened options opportunities, hedging rules and funded-account constraints.

The constant-full rule earns positive historical profit above cash. Learned sizing raises the funded point estimate, with a 0.208 Sharpe gain and approximately $47,800 additional profit, but its gain interval includes zero. It also admits more trades, 110 versus 75, under the same account rules. Annual results vary, including a loss relative to cash in 2024. The evidence supports a higher aggregate historical estimate for learned sizing, with uncertainty about its repeatability.

The comparison uses previously inspected history and differs across training, selection and evaluation in capital, costs and admission. Execution and option exercise are simplified. These results establish neither repeatable learning benefits nor live profitability or capacity. Most account growth is cash interest; trading value is measured above cash.

# A Signal and learning specification

Choose the expiry closest to 30 days in the historical anchor selection, then the put in that expiry nearest absolute delta 0.25, within 0.075. Ties favor narrower spreads, higher open interest and lower contract identifier. Entry requires standard 100-share contracts, positive noncrossed quotes, open interest at least 100, width at most 25% of midpoint and 7–55 days remaining. The IV rank includes the signal close; relatedness is fixed monthly from the prior 126 sessions. Profitability memory uses completion dates, not entry dates.

Admission stress sums each name's worst modeled option-plus-hedge loss across spot shocks of $-20/0/+20\%$ and IV shocks of $-10/0/+10$ percentage points, without credit for pair correlation. Stress loss is capped at 5% of budget; reserve (gross option premium plus half netted hedge notional) at 80%, hedge notional at twice budget and gross vega at \$20,000 per IV point per \$1 million. Global checks use account equity; pair checks use \$1 million scaled by current/initial account equity. Proposals are processed in pair-identifier order. These are research limits, not broker margin. Corporate events or unsupported declared distributions exit affected pairs; missing entry quotes reject proposals, while held marks may use last-known IV with current spot, rates and known dividends. Expiry is cash-settled with a European valuation approximation; market impact, early assignment and forced liquidation are not fully modeled.

The tree has up to eight leaves, with minimum leaf count one eighth of its training sample rounded down, bounded between 8 and 63. Fewer than 16 examples use full sizing. CEM begins uniformly each month and mixes 30% current probability with 70% elite frequency, multiplies by 0.97 and adds 0.03 equally across the three actions. The random seed is 2026.

## A.1 Inputs available at the signal date

The model uses the following 15 numerical inputs. Proposed-direction quantities are oriented as long leg minus short leg. IV and volatility levels are decimal annualized values. Trailing realized variance is 252 times mean squared daily simple returns over the latest 20 listed trading sessions, with at least 10 observations, including the signal-close return. Returns generally exclude dividends; adjusted-close percentage changes provide the fallback series.

| Input | Definition |
|---|---|
| Mean IV | Mean of the two selected-contract IVs. |
| Oriented IV gap | Long-minus-short IV divided by mean IV. |
| Rank strength | Twice the absolute distance of the IV-gap percentile from 0.5. |
| Oriented variance-premium gap | Long-minus-short difference in IV squared minus trailing realized variance, divided by mean IV squared. |
| Mean RV | Mean of the square roots of the two trailing realized variances. |
| Beta; correlation | Two inputs: 126-session A-on-B beta and correlation, fixed at monthly formation. |
| Mean expiry; expiry gap | Two inputs: mean calendar days to expiry and long-minus-short expiry days. |
| Commission in IV points | Sum across legs of \$0.65 divided by each contract's dollar vega per IV point. |
| VIX | Most recent strictly prior VIX close divided by 100. |
| 91-day profit; count | Two inputs: completed full-size profit per initial vega averaged over the trailing 91 calendar days and the number of completed outcomes. The profit input is divided by 100 as a fixed scale. |
| 365-day profit; count | Two inputs: the same measurements over 365 calendar days. |

The first three inputs are core fields. Each of the other 12 has a binary availability flag, giving 27 inputs in total. Nonfinite optional values receive zero and flag zero; observed values have flag one. Profit features use only outcomes completed before the signal date; future returns and profits enter outcomes, not inputs.

# B Performance definitions and evidence

Let $N_t$ be account equity and $K_t$ all-cash benchmark wealth at close $t$. Both start at \$10 million in January 2018. The benchmark compounds the previous available WRDS zero-curve rate nearest 30 days over calendar days using a 365-day year. Daily excess return and annualized Sharpe are

$$e_t = \left(\frac{N_t}{N_{t-1}} - 1\right) - \left(\frac{K_t}{K_{t-1}} - 1\right), \qquad \widehat{S} = \sqrt{252}\,\bar{e}/s_e.$$

Here $\bar{e}$ and $s_e$ are the sample mean and standard deviation of daily excess returns; 252 is assumed trading days per year. For a reporting period with preceding close 0 and final close $T$, dollar profit above cash is $(N_T - K_T) - (N_0 - K_0)$. Set $Q_t = N_t/K_t$: annualized relative growth is $(Q_T/Q_0)^{1/Y} - 1$, where $Y$ is elapsed days divided by 365.25. Cash-relative drawdown is the largest percentage decline in $Q_t$ from its running peak within that period, including close 0.

Annual-table account return is $N_T/N_0 - 1$; above-cash percentage points subtract benchmark return $K_T/K_0 - 1$. Period returns are unannualized, including partial 2025; Sharpe remains annualized. At the 31 December 2021 boundary, learner and constant-full equity are \$10.568 million and \$10.581 million, with four and three open trades respectively. Benchmark wealth is \$10.510 million. Chart curves subtract each policy's opening dollar excess wealth; accounts and holdings are not reset.

The gain interval resamples the learner and constant-full daily excess returns with identical indices: 1,000 samples concatenate randomly started 126-session blocks, wrap at the series end and truncate to original length. Each sample recalculates the Sharpe difference; endpoints are its 2.5th and 97.5th percentiles. Fitted policies are fixed and research selection is unadjusted.

**Evidence sources.** Results and implementation use repository snapshot 688bc39. The directory `reports/p25_validation_split` contains `periods.csv`, `annual.csv`, `holdout_intervals.csv` and `selection.json`. Figure 3 uses the selected account in `data/p25_validation_split_experiment.zip` and the constant-full account under `data/submission/artifacts/contracts/p25_capital_sensitivity`. The complete 125-pair roster, candidate-date ranges and proposal counts are retained in the companion `paper/pair_universe.csv`; ranges describe recorded candidates, not continuous quote coverage or executed trading dates. Inherited formation calendars and usable contracts condition the universe. Vendor data retain their institutional license restrictions.

# References


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